Selective Context: Prompt Compression for Efficient LLM-Based News Understanding

Efficient News Understanding Through Selective Context Compression

PEGAI Course Project (PES1UG24AM163 / PES1UG24AM142)

WHY - The problem

Large language models (LLMs) can process long news articles, but supplying an entire article for every task can be inefficient. Long contexts increase the amount of text the model must process, which can raise inference cost and latency. At the same time, simply shortening an article risks removing facts, entities, relationships, and details that are important to the task. The challenge is therefore not just to make context shorter, but to reduce unnecessary information while preserving the content a model needs to produce useful outputs.

WHAT - The proposed solution

Selective Context is a context-compression approach that identifies and removes less informative phrases from a source text while retaining phrases considered more important. In this project, I propose to apply a phrase-level, self-information-based selection process to BBC News articles, then provide the resulting compressed context to language models for downstream news-processing tasks. The aim is to examine whether a shorter context can still support reliable task performance.

How the solution is explored

The project compares an uncompressed baseline with five configured phrase-removal levels: 20%, 35%, 50%, 65% and 80%. The same compression pipeline is examined across four tasks:

- News summarisation - generate a concise summary from the article.
- Question answering - answer factual questions using the article context.
- Response generation - produce context-grounded responses in a short dialogue setting.
- Original context reconstruction - attempt to recover the source article from its compressed version.

What success means

The solution will be assessed by comparing task outputs across compression levels against an uncompressed baseline. Evaluation uses task-appropriate measures, including ROUGE and BERTScore for text similarity, Exact Match and token F1 for question answering, and BLEU and METEOR for reconstruction. Context reduction is considered alongside output quality: a useful setting should reduce the input while retaining acceptable performance for the task. The project does not assume that one compression level will work equally well for every task.

Expected outcome

The expected outcome is an experimental assessment of the trade-off between context length and downstream LLM performance in news understanding. By studying several tasks under the same compression settings, the project aims to identify where selective phrase removal is useful, where it causes information loss, and how the appropriate compression level may depend on the task.

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